Sleep-Consolidated Language Modeling

LeMON Lab
Learning, Meaning, & Natural Language

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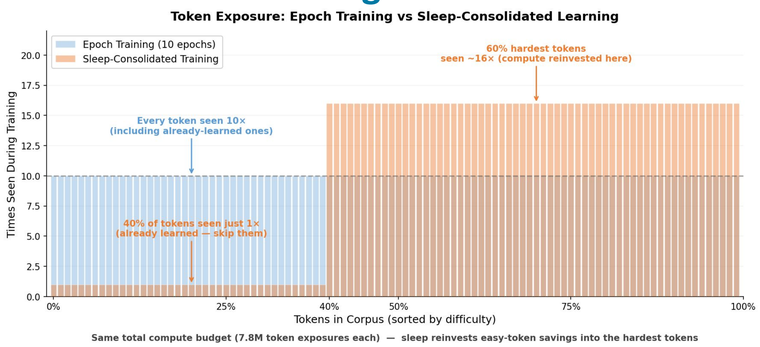
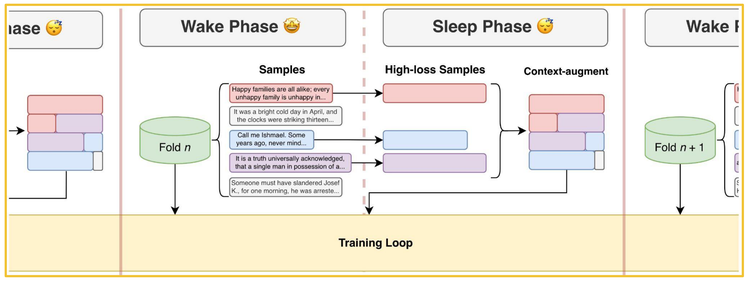
Background



- Standard LM training is expensive[4]
  - Much less data-efficient than humans
  - Requires significant compute
- Epoch training schedules aren't cognitively plausible
  - Humans live through time linearly
  - Replay happens during sleep[3]

Methods: Training Schedule



Goal: Replace epoch schedule with sleep-like, cognitively-plausible replay schedule

Sleep Mechanism

- Incorporates biologically-inspired replay phase into LLM training
- Model only sees each data point once during wake
- Wake samples selected for sleep phase based on difficulty
- High-loss samples are "recontextualized" to encourage abstraction

Linguistic Evaluation

- **BLiMP**: Grammar and linguistic understanding
- **WUGS**: Derivational morphology
- **Entity Tracking:** Discourse entity states

Compare Sleep LMs to Baseline (Epoch-schedule LMs)

Methods: Dataset and Model

Dataset: BabyLM 100M Corpus[1]

- Approximates linguistic input a child receives by age 13
- Combination of sources from 2 domains: transcribed speech & child-directed text

Model: RoBERTa-Small[2]

- 8 hidden layers, 8 attention heads
- Hidden size: 256, intermediate size: 2048
- Trained on masked language modeling (MLM)

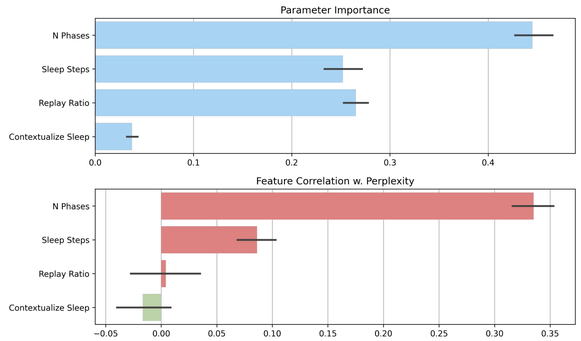
Results

Baseline PPL: 36.63

Hyperparameter Sweep

Best PPL: 109.133

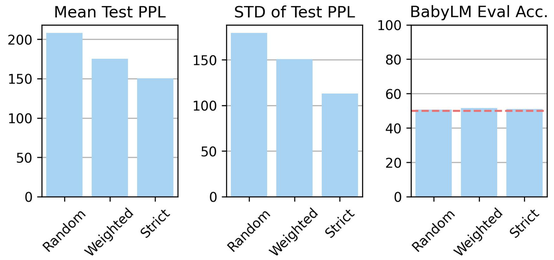
- Gridsearch over # phases, ratio, contextualize, steps
- Fewer phases = Better
  - Longer wake phases
  - Less overfitting(?)
- Replay Ratio and Contextualization do not have significant corr.



Replay Strategy

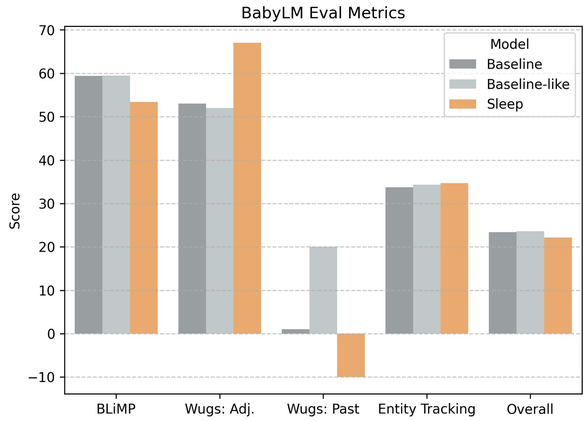
Best PPL: 150.21

- Tested selection of samples for buffer
- Strict strategy was best
  - Mean and STD of PPL reduced
- Similar in BabyLM eval



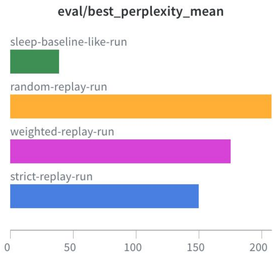
BabyLM Evaluation

- Sleep LM > Baseline in Adj. Nominalization
  - More human-like than baseline
- Linguistic understanding (BLiMP) is still poorer than baselines
- Entity tracking results are inconclusive

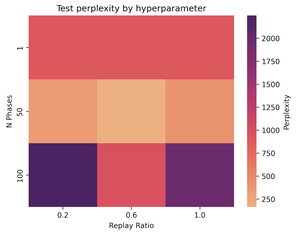


Conclusions

Currently, standard epoch training outperforms sleep mechanism in both perplexity and BabyLM evaluation metrics.



- Selective replay reduces wasted compute on already-mastered data
  - Strict replay = more consistency
- "Catastrophic forgetting" may occur
  - PPL drops drastically
  - Plasticity decay may further close the gap between biological and LM language acquisition
- Promising results in hyperparameter sweeps
  - perplexity seems to be lower when diverging from epoch schedule



Limitations & Future Work

- Still underperforming epoch-training models
  - More rigorous/comprehensive hyperparameter tuning
- Replay selection strategy - loss might be encoding noise
  - Try other ways of selecting useful samples for learning
- Implement plasticity decay: weight downscaling + hierarchical layer freezing
  - Could help model retain information from earlier folds
- Extend to larger-scale models and datasets beyond BabyLM
  - Looking at PPL pattern with other datasets

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References

[1] Hu et. al., 2024. "Findings of the Second BabyLM Challenge: Sample-Efficient Pretraining on Developmentally Plausible Corpora."
[2] Liu et. al., 2019. "RoBERTa: A Robustly Optimized BERT Pretraining Approach."
[3] Rasch and Born. 2013. "About Sleep's Role in Memory."
[4] Warstadt and Bowman. 2024. "What Artificial Neural Networks Can Tell Us About Human Language Acquisition."

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https://lemn-lab.github.io/datascience.ucsd.edu
https://lasyayadlapati.github.io/sleep-consolidated-learning/